

Author

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I am passionate about developing web applications that serve real purpose.

Description

A complete web-based Hospital Management System that enables Admins, Doctors, and Patients to interact with the system seamlessly.

The project is built using the following stack Flask (API), VueJS (UI), Bootstrap (UI styling), SQLite (database), Redis (async jobs).

Technologies used

Flask + SQLAlchemy :

Managing ORM operations and creating HTTP endpoints.

VueJS :

Used to build a responsive and modular front-end. Vue Router handles multi-role navigation across dashboards.

Bootstrap :

For layouts, components, forms, tables, grids, and overall visual structure and both for caching frequently accessed data (doctor lists, departments, availability)

Redis : Used both for catching frequently accessed data (doctor lists, departments,

availability) and powering Celery workers.

Celery : Daily appointments remainders, Monthly doctor activity reports and CSV Report.

DB Schema Design

The database schema is designed to manage the core entities of the hospital system, with models dedicated to storing and differentiating the three required user roles (Admin, Doctor, Patient).

Stores details regarding scheduled meetings, including the Patient ID, Doctor ID, Date, Time, and the current Status.

Tables collectively manage authentication, scheduling, medical records, and hospital operations with clean, scalable relationships.

API Design

I have not used any API services that were integrated during the development.

Architecture and Features

The application is structured around the three distinct user roles, each accessing a dedicated dashboard for role-specific functions.

Displays critical metrics: Total number of doctors, patients, and appointments.

The application uses a modular MVC-style backend where models, services, and controllers are cleanly separated, ensuring maintainability and scalable feature expansion. Mainly separated by the resources and services.

Core features include authentication, appointment management, treatment logging, automated reminders, monthly reports, and CSV exports operating seamlessly across all roles.